CFS RFID | Electrical Wiring Reference

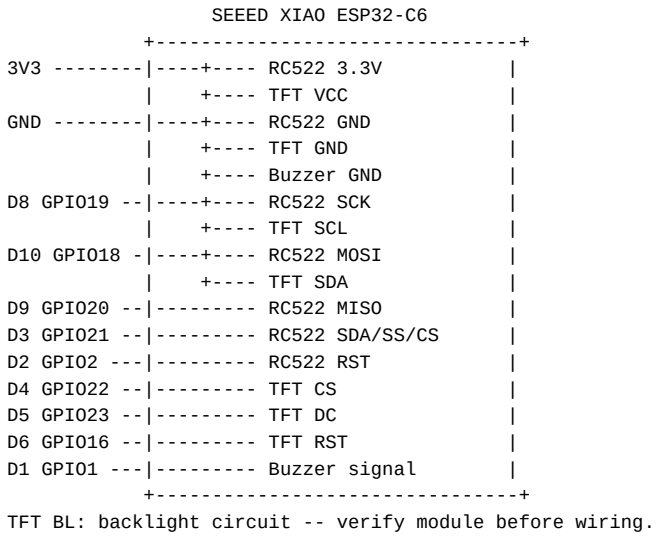
Seeed XIAO ESP32-C6 + RC522 + ST7789 1.54-inch TFT + buzzer

Firmware reference: v0.3.7 TEST | 26 October 2026 | 3.3 V logic

1. Complete connection schedule

Controller pin	Connects to	Function
XIAO 3V3	RC522 3.3V; TFT VCC	3.3 V power rail
XIAO GND	RC522 GND; TFT GND; buzzer GND	Common ground
D8 / GPIO19	RC522 SCK; TFT SCL	Shared SPI clock
D10 / GPIO18	RC522 MOSI; TFT SDA	Shared SPI data out
D9 / GPIO20	RC522 MISO	SPI data in
D3 / GPIO21	RC522 SDA / SS / CS	RC522 chip select
D2 / GPIO2	RC522 RST	Reader reset
D4 / GPIO22	TFT CS	Display chip select
D5 / GPIO23	TFT DC	Display data/command
D6 / GPIO16	TFT RST	Display reset
D1 / GPIO1	Buzzer/speaker signal	Existing signal output
GPIO3; GPIO14	No external connection	Reserved antenna RF switch

2. Shared-SPI wiring schematic



3. Electrical and integration notes

TFT header order. As marked on the photographed ST7789 PCB: GND, VCC, SCL, SDA, RST, DC, CS, BL. "SDA" here means SPI MOSI, not I2C SDA.

TFT backlight. The test arrangement discussed uses BL at 3V3, but the backlight resistor/current-limit implementation on this exact module is unverified. Check the module circuit and backlight current before permanent direct connection; use an appropriate current-limited driver if necessary.

Shared SPI. Both devices share SCK and MOSI. Only the RC522 uses MISO. Each device has a separate CS line; deselect both before starting SPI transactions.

Antenna selection. Firmware selects XIAO onboard ceramic Wi-Fi antenna (external U.FL connector damaged). GPIO3 and GPIO14 are dedicated to the XIAO RF antenna switch and must not be used for peripherals.

Power. Power the RC522 from 3.3V, not 5V. Check total TFT + RC522 + buzzer load against the XIAO 3.3V supply capability. Use common ground.

RFID pins. The RC522 pin marked SDA is its SPI chip-select (SS/CS) in this project. RC522 IRQ is deliberately unconnected.

Buzzer. D1/GPIO1 is signal only; buzzer power wiring depends on its type. Do not connect a high-current speaker directly to a GPIO. Use a transistor/driver where required.

Deployment protection. This is a wiring reference only. Existing printer SSH/SCP sync, database and CFS logic should remain unchanged.

Validation. The firmware pin mapping was checked against v0.3.7 TEST source. Physical solder joints and backlight circuitry cannot be verified from firmware.

4. Firmware pins (source of truth)

```
RFID_SCK_PIN=19   RFID_MISO_PIN=20   RFID_MOSI_PIN=18
RFID_SS_PIN=21    RFID_RST_PIN=2    SPK_PIN=1
TFT_CS_PIN=22     TFT_DC_PIN=23     TFT_RST_PIN=16
ANT_SWITCH_POWER_PIN=3       ANT_SWITCH_SELECT_PIN=14
```